Electronics and Electrical Communications Engineering Department

Faculty of Engineering

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Neural Networks

Assignment 1
Classical Machine Learning Methods

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Contents

1 Introduction **3**

2 Pipeline Overview **3**

3 Feature Extraction **4**

4 K-Means Clustering **4**

5 SVM and Active Learning **5**

6 Results **5**
6.1 Accuracy 5
6.2 Iterations 6

7 Manual Effort **6**

8 Conclusion **6**

9 Pipeline 2: Manual Seed + Augmentation + Self-Training **6**
9.1 Pipeline Diagram 7
9.2 Final Results 8
9.3 Iteration Progress 8
9.4 Visualization 8
9.5 Manual Effort 9
9.6 Comparison with Pipeline 1 9
9.7 Discussion 10

List of Figures

1 Pipeline 1: K-Means + SVM + Active Learning 3
2 Example of HOG Feature Representation 4
3 K vs Accuracy Before SVM 4
4 Accuracy Before vs After SVM 5
5 Pipeline 2: Semi-Supervised Learning Framework 7
6 Initial Human Seed Samples (10 per class) 8
7 Example of Data Augmentation (7 variants) 9
8 Hard Samples (Iteration 1) 9
9 Hard Samples (Iteration 2) 9

1 Introduction

This report presents a semi-supervised pipeline for labeling a dataset of 10,000 handwritten digit images. The objective is to achieve high classification accuracy while minimizing manual labeling effort by combining clustering, feature extraction, and active learning.

2 Pipeline Overview

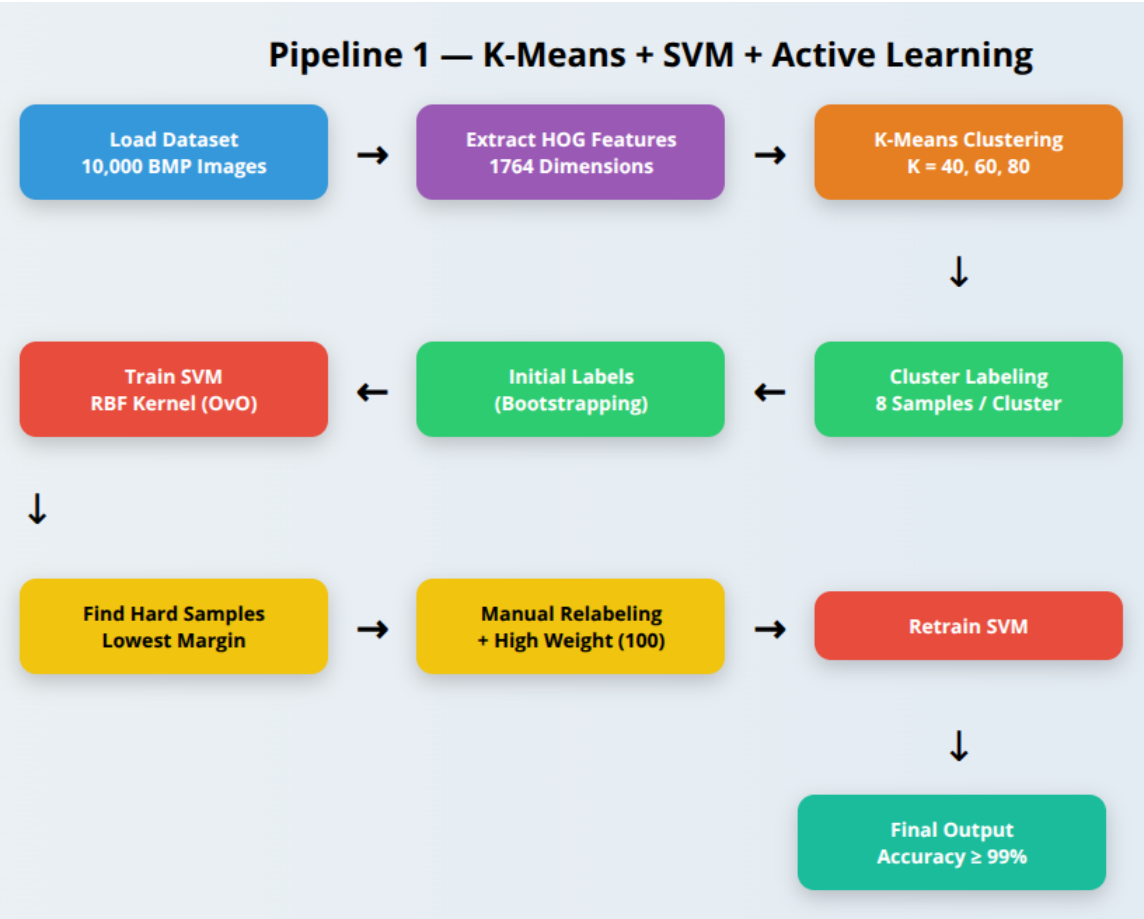

Figure 1: Pipeline 1: K-Means + SVM + Active Learning

The pipeline consists of the following steps:

- Load dataset (10,000 BMP images)
- Extract HOG features
- Apply K-Means clustering
- Label clusters (bootstrapping)
- Train SVM classifier
- Apply active learning for refinement

3 Feature Extraction

Histogram of Oriented Gradients (HOG) was used to extract features from images.

- Image size: 32×32
- Feature vector size: 1764

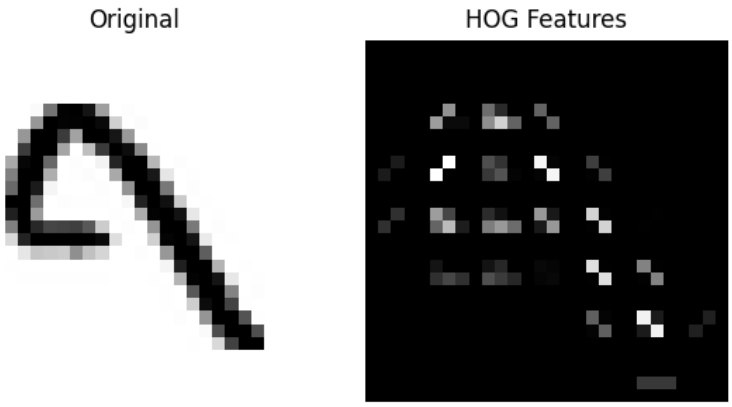

Figure 2: Example of HOG Feature Representation

4 K-Means Clustering

K-Means clustering was applied with different values of K:

- K = 40
- K = 60
- K = 80

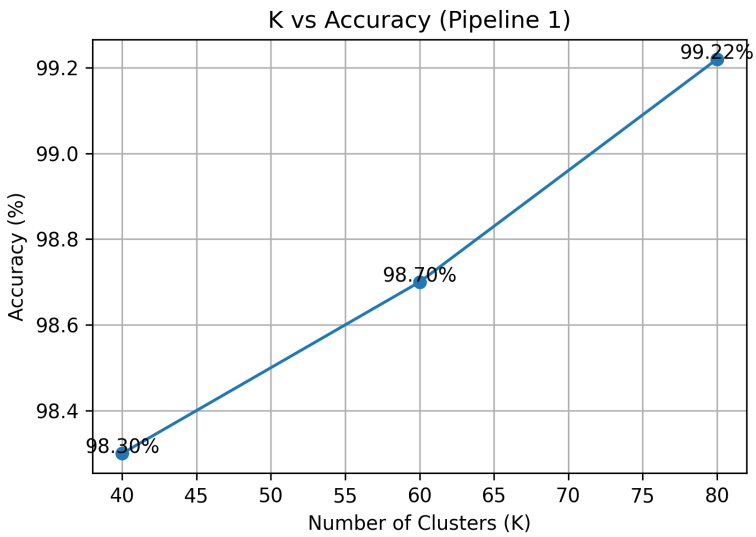

Figure 3: K vs Accuracy Before SVM

5 SVM and Active Learning

A multi-class SVM with RBF kernel (One-vs-One strategy) was used.
Active learning was applied as follows:

- Select 30 least confident samples (lowest margin)
- Relabel manually
- Assign weight = 100
- Retrain SVM

The process stops when:

- Accuracy $\geq 99\%$
- Improvement $< 0.1\%$

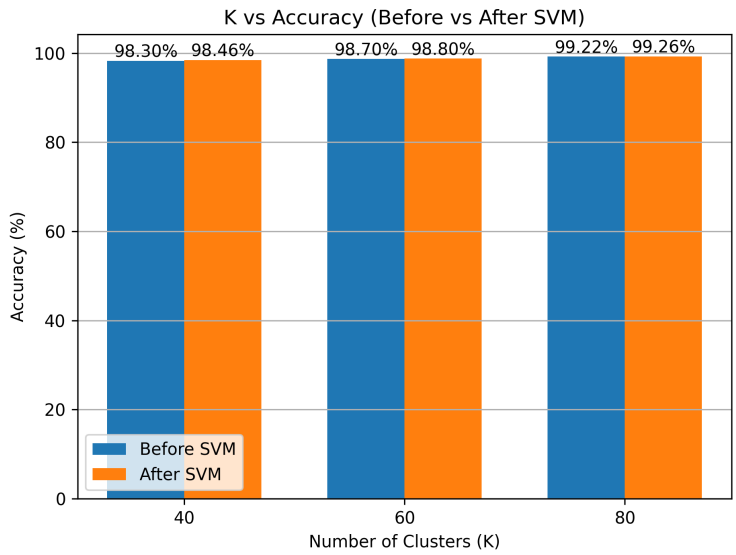

Figure 4: Accuracy Before vs After SVM

6 Results

6.1 Accuracy

K	Before SVM (%)	After SVM (%)
40	98.30	98.46
60	98.70	98.80
80	99.22	99.26

Table 1: Accuracy Comparison

6.2 Iterations

K	Iterations
40	2
60	2
80	2

Table 2: Number of Iterations

7 Manual Effort

For K = 80:

- Cluster labeling: $80 \times 8 = 640$ images
- Active learning: $30 \times 2 = 60$ images

Total images manually handled:

$$700 \text{ images}$$

Total time:

- Cluster labeling: 1600 seconds
- Active learning: 600 seconds

$$\text{Total time} = 2200 \text{ seconds} \approx 36.7 \text{ minutes}$$

8 Conclusion

The proposed pipeline significantly reduces manual labeling effort from 27.8 hours to approximately 37 minutes while achieving 99.26% accuracy.

K-Means clustering provides strong initial labels, while SVM and active learning refine ambiguous samples. The improvement from SVM is more noticeable at lower K values, while higher K values already produce highly pure clusters.

9 Pipeline 2: Manual Seed + Augmentation + Self-Training

This pipeline adopts a semi-supervised learning approach starting from a small, high-quality human-labelled seed set. The model progressively expands its knowledge using data augmentation, active learning, and self-training.

9.1 Pipeline Diagram

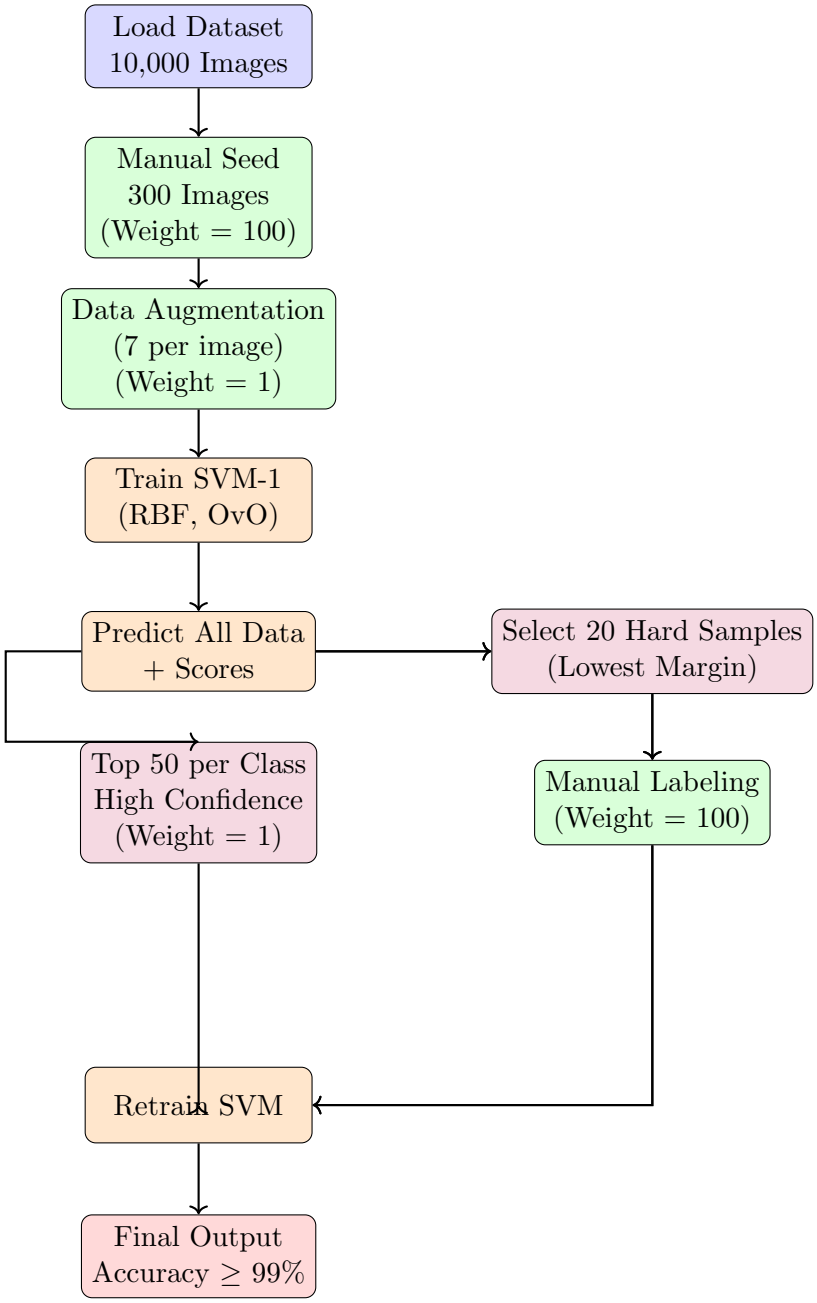

Figure 5: Pipeline 2: Semi-Supervised Learning Framework

9.2 Final Results

Metric	Value
Final Accuracy	98.87%
Iterations	2
Manual Labels	340
Pseudo Labels	1000
Estimated Time	56.67 minutes

Table 3: Pipeline 2 Performance Summary

9.3 Iteration Progress

Detailed iteration results are stored in the generated CSV file:
`pipeline2_iterations.csv` :contentReference[oaicite:0]index=0
Each iteration records:

- Accuracy improvement
- Number of pseudo-labelled samples
- Number of manually labelled samples

9.4 Visualization

Figure 6: Initial Human Seed Samples (10 per class)

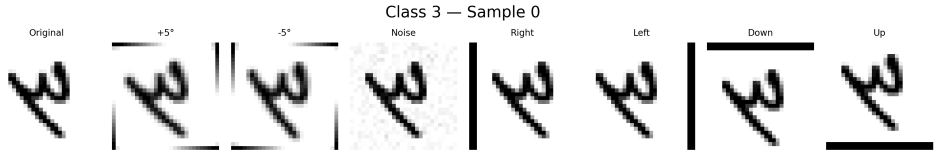

Figure 7: Example of Data Augmentation (7 variants)

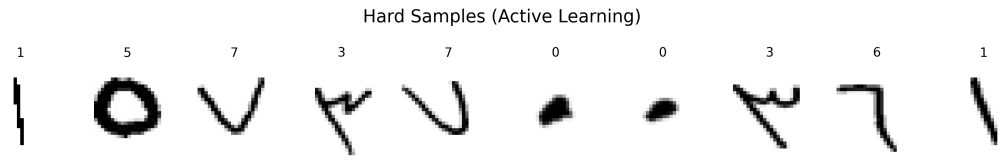

Figure 8: Hard Samples (Iteration 1)

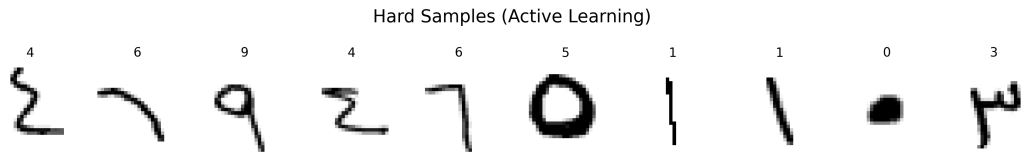

Figure 9: Hard Samples (Iteration 2)

9.5 Manual Effort

Manual labeling consists of:

- Initial seed: 300 images
- Active learning: $20 \times 2 = 40$ images

Total manual labels:

$$300 + 40 = 340$$

Estimated time:

$$340 \times 10 = 3400 \text{ seconds} \approx 56.67 \text{ minutes}$$

9.6 Comparison with Pipeline 1

Metric	Pipeline 1	Pipeline 2
Accuracy	99.26%	98.87%
Manual Images	700	340
Manual Time	36.7 min	56.7 min
Iterations	2	2

Table 4: Pipeline Comparison

Pipeline 1 achieves higher accuracy due to clustering-based initialization, while Pipeline 2 significantly reduces manual labeling effort by leveraging self-training. However, Pipeline 2 requires slightly more total time due to per-image labeling cost.

9.7 Discussion

Pipeline 2 demonstrates the effectiveness of semi-supervised learning. Starting from a small, high-quality seed set, the model successfully expands its training data using both active learning and self-training.

Although the final accuracy (98.87%) is slightly below Pipeline 1, the reduction in manual labeling effort is significant. This pipeline better reflects real-world scenarios where labeled data is scarce and expensive.

The combination of:

- Human supervision (high confidence)
- Model confidence (self-training)
- Iterative refinement

creates a scalable and efficient labeling strategy.